

1. Introduction

In this module, I explored the Domain Name System (DNS), a core component of how the Internet functions. Through a series of tasks, I learned what DNS is, how domain names are structured hierarchically, the different types of DNS records, and how DNS requests are made and resolved. This gave me a practical understanding of how domain names are translated into IP addresses and how this process supports everyday internet use.

2. Module Sections

What does DNS stand for?

What is DNS?

DNS (Domain Name System) provides a simple way for us to communicate with devices on the internet without remembering complex numbers. Much like every house has a unique address for sending mail directly to it, every computer on the internet has its own unique address to communicate with it called an IP address. An IP address looks like the following 104.26.10.229, 4 sets of digits ranging from 0 - 255 separated by a period. When you want to visit a website, it's not exactly convenient to remember this complicated set of numbers, and that's where DNS can help. So instead of remembering 104.26.10.229, you can remember tryhackme.com instead.

Woop woop! Your answer is correct

Answer the questions below

What does DNS stand for?

Domain Name System

Correct Answer

Task 2 Domain Hierarchy

Task 3 Record Types

What is the maximum length of a subdomain?

63

Which of the following characters cannot be used in a subdomain (3 b _ -)?

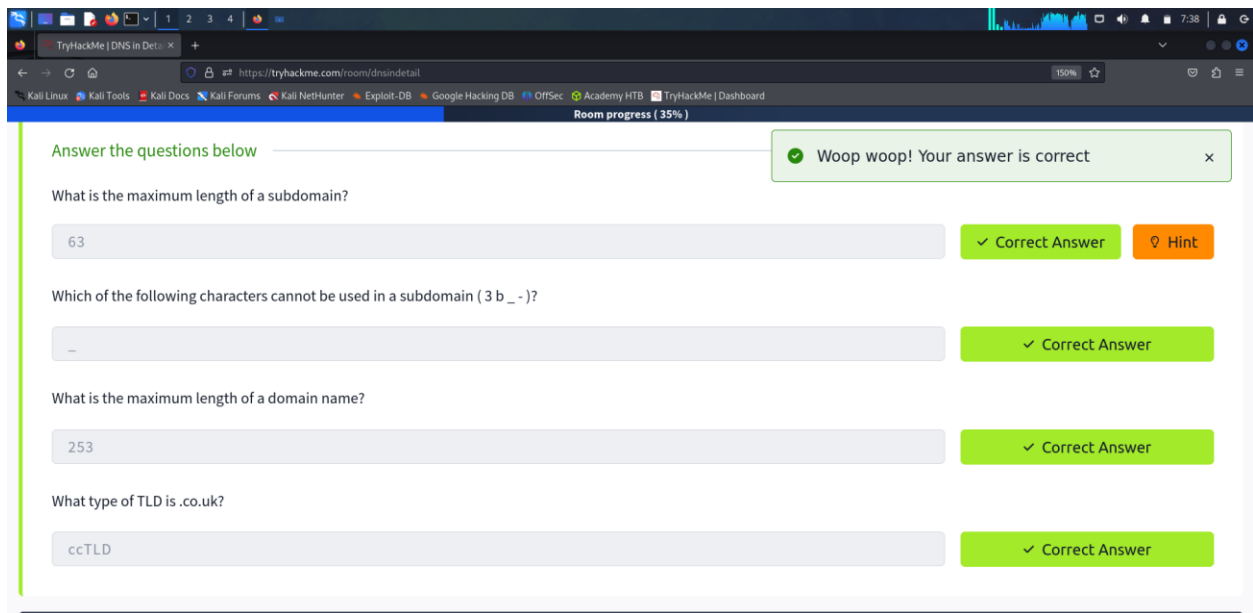
—

What is the maximum length of a domain name?

253

What type of TLD is .co.uk?

ccTLD

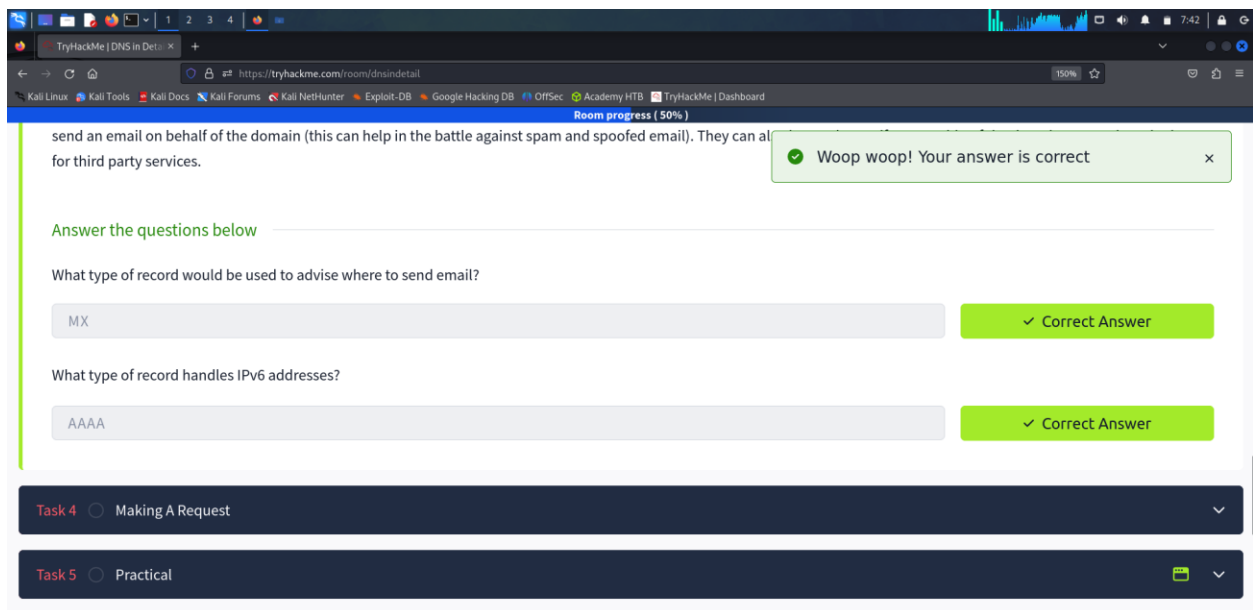


What type of record would be used to advise where to send email?

MX

What type of record handles IPv6 addresses?

AAAA



What field specifies how long a DNS record should be cached for?

TTL

What type of DNS Server is usually provided by your ISP?

Recursive

What type of server holds all the records for a domain?

Authoritative

The screenshot shows a web browser window with the URL `https://tryhackme.com/room/dnsdetail`. The browser's address bar and tabs are visible at the top. Below the browser window, a notification box says "Woop woop! Your answer is correct". The main content area is titled "Answer the questions below" and contains three questions, each with a text input field and a green "Correct Answer" button.

the response should be saved for locally until you have to look it up again. Caching saves on having to ma

Woop woop! Your answer is correct

Answer the questions below

What field specifies how long a DNS record should be cached for?

TTL

✓ Correct Answer

What type of DNS Server is usually provided by your ISP?

recursive

✓ Correct Answer

What type of server holds all the records for a domain?

authoritative

✓ Correct Answer

Task 5 Practical

I did the practical and found the answers as shown below

The screenshot shows a web browser window with the URL `https://tryhackme.com/room/dnsindetail`. The page is titled "Task 5" and "Practical". It contains a description of DNS queries and a "View Site" button. Below the description, there are two questions: "What is the CNAME of shop.website.thm?" and "What is the value of the TXT record of website.thm?". The first question has a text input field containing `shops.myshopify.com` and a "Correct Answer" button. The second question has a text input field containing `{ }`. To the right of the task, there is a terminal window showing the output of `nslookup --type=CNAME shop.website.thm` and `nslookup website.thm`. The terminal output shows the canonical name `shops.myshopify.com` and the TXT record value `{ }`.

Room progress (76%)

Task 5 Practical

Using the website on the right, we can build requests to make DNS queries and view the results. The website will also show you the command you'd need to run on your own computer if you wished to make the requests yourself. [View Site](#)

Answer the questions below

What is the CNAME of shop.website.thm?

`shops.myshopify.com`

✓ Correct Answer ⚠ Hint

What is the value of the TXT record of website.thm?

`{ }`

DNS Type: subdomain [Send DNS Request](#)

```
user@thm:~$ nslookup --type=CNAME shop.website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
shop.website.thm canonical name = shops.myshopify.com

user@thm:~$ nslookup website.thm
```

How DNS Works

TryHackMe | DNS in Detail

https://tryhackme.com/room/dnsindetail

Room progress (85%)

the requests yourself.

Answer the questions below

What is the CNAME of shop.website.thm?

shops.myshopify.com

Correct Answer

Hint

What is the value of the TXT record of website.thm?

THM{7012BBA60997F35A9516C2E16D2944FF}

Correct Answer

Hint

What is the numerical priority value for the MX record?

How DNS Works

Woop woop! Your answer is correct

user@thm:~\$ nslookup --type=CNAME shop.website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
shop.website.thm canonical name = shops.myshopify.com

user@thm:~\$ nslookup --type=TXT website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
website.thm text = "THM{7012BBA60997F35A9516C2E16D2944FF}"

user@thm:~\$ nslookup website.thm

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https://tryhackme.com/room/dnsindetail

Room progress (92%)

THM{7012BBA60997F35A9516C2E16D2944FF}

Correct Answer

Hint

What is the numerical priority value for the MX record?

30

Correct Answer

Hint

What is the IP address for the A record of www.website.thm?

Submit

Hint

How likely are you to recommend this room to others?

Woop woop! Your answer is correct

Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
shop.website.thm canonical name = shops.myshopify.com

user@thm:~\$ nslookup --type=TXT website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

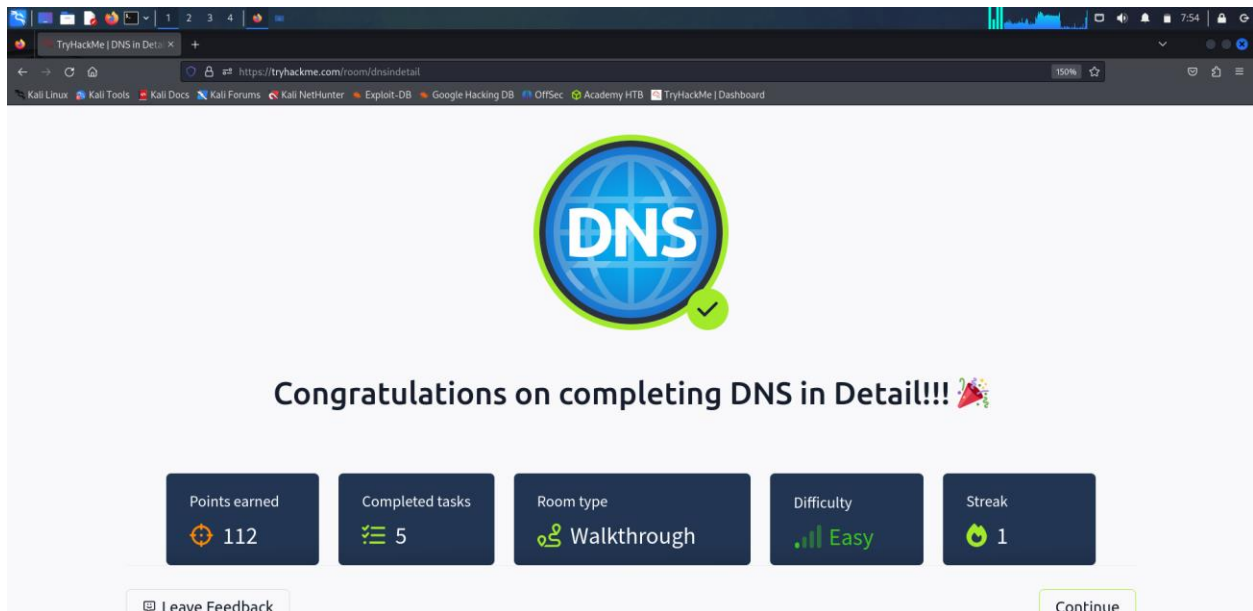
Non-authoritative answer:
website.thm text = "THM{7012BBA60997F35A9516C2E16D2944FF}"

user@thm:~\$ nslookup --type=MX website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
website.thm mail exchanger = 30 alt4.aspmx.l.google.com

user@thm:~\$ nslookup website.thm

3. Shareable Link and Final Screenshot



4. Social Media Posting

https://x.com/intent/post?url=https%3A%2F%2Ftryhackme.com%2Froom%2Fdnsindetail%3Futm_source%3Dtwitter%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom&text=Share%20your%20achievement&via=realtryhackme&hashtags=tryhackme

https://www.facebook.com/sharer/sharer.php?u=https%3A%2F%2Ftryhackme.com%2Froom%2Fdnsindetail%3Futm_source%3Dfacebook%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom&t=Share%20your%20achievement

https://www.linkedin.com/sharing/share-offsite/?url=https%3A%2F%2Ftryhackme.com%2Froom%2Fdnsindetail%3Futm_source%3Dlinkedin%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom

5. Conclusion

This module helped me see the DNS not just as a technical concept but as a vital service that underpins the accessibility and efficiency of the web. I now understand the role of root, top-level, and subdomains in the domain hierarchy, and I can identify various DNS record types like A, AAAA, MX, and CNAME. Most importantly, learning how DNS queries work—from a user's request to the resolution process—highlighted the complexity and importance of DNS in cybersecurity. It also made me more aware of potential vulnerabilities within DNS and how attackers might exploit them, reinforcing the importance of securing this critical system.